# Regulatory, Clinical, and Legal Landscapes of Sodium Valproate: A Comparative Analysis of Ireland, England, and France

## 1. Introduction to the Sodium Valproate Crisis

Sodium valproate (VPA), synthesized initially in 1882 as a laboratory solvent, was discovered to possess potent antiepileptic properties in France in the 1960s.[1] First introduced to the European market in 1967 under the brand name Depakine by Sanofi (then Laboratoires Berthier), the chemical rapidly established itself as a cornerstone of neurological and psychiatric pharmacotherapy.[2] Marketed variously as Epilim, Depakote, Convulex, and Episenta, sodium valproate is a broad-spectrum antiepileptic drug (AED) that has demonstrated unparalleled efficacy in the management of generalized and focal epilepsies, as well as acting as a potent mood stabilizer for the treatment of manic episodes associated with bipolar affective disorder.[4] For decades, it has been globally recognized as an essential medicine, retaining a prominent position on the World Health Organization's Model List of Essential Medicines due to its life-saving capacity to prevent sudden unexpected death in epilepsy (SUDEP).[7]

However, the clinical triumph of sodium valproate is irrevocably marred by a catastrophic safety profile regarding its teratogenicity. The widespread prescription of this medication to women of childbearing potential has precipitated one of the most profound and devastating iatrogenic public health crises in modern medical history. Clinical and epidemiological data now uniformly indicate that in utero exposure to sodium valproate induces a severe spectrum of morphological anomalies and cognitive pathologies, collectively designated as Fetal Valproate Spectrum Disorder (FVSD).[9] Infants exposed to the drug during gestation face an approximate 10% risk of major congenital malformations and an alarming 30% to 40% risk of severe, lifelong neurodevelopmental disorders.[11] Despite emerging evidence of these risks appearing in medical literature as early as the 1970s and 1980s, the pharmaceutical industry and national health regulators engaged in systemic inertia, allowing tens of thousands of pregnancies across Europe to be exposed to the chemical without adequate informed consent or risk mitigation.[1]

The ensuing fallout has generated disparate regulatory, legal, and investigatory responses across Europe. The promptness and efficacy with which states have addressed this crisis vary drastically. France and England mobilized national inquiries and legal frameworks relatively rapidly—addressing the escalating scandal with the urgency of extinguishing a rapidly spreading fire.[15] In stark contrast, the Republic of Ireland has been characterized by profound

bureaucratic delays, only recently commencing a non-statutory inquiry in July 2025, years after initial governmental commitments were made.[17] Concurrently, the regulatory landscape is undergoing a paradigm shift. Recent pharmacokinetic data highlighting the risks of paternal exposure have prompted sweeping new restrictions across the United Kingdom and Europe, applying stringent prescribing protocols to both males and females under the age of 55.[20]
As the prescription of sodium valproate faces mounting restrictions, manufacturers have historically pivoted their commercial strategies to maintain production volumes, pushing the drug toward male demographics and off-label psychiatric indications in what has been critically analyzed as a "factory excess" marketing paradigm.[15] Simultaneously, the therapeutic vacuum created by valproate restrictions is increasingly being filled by alternative treatments, notably medical cannabis and highly purified cannabidiol (CBD) isolates. This shift introduces complex clinical and economic dynamics, contrasting the low-cost, highly accessible, genericized distribution networks of valproate against the heavily regulated, premium-priced emerging cannabinoid markets.[24] This comprehensive report provides an exhaustive analysis of the legalities, investigatory timelines, teratological impact, and market dynamics surrounding sodium valproate in Ireland, England, and France, evaluating the transition toward alternative therapeutic pathways in contemporary neurological care.

# 2. Teratogenic Profiling and the Pathogenesis of Fetal Valproate Spectrum Disorder

The teratogenicity of sodium valproate is unparalleled among contemporary antiepileptic medications. While older generation AEDs such as phenobarbital and phenytoin carry documented risks, the developmental devastation wrought by valproate exposure during embryogenesis is systemic, affecting multiple organ systems and the fundamental cellular architecture of the central nervous system.[26] The resulting condition, Fetal Valproate Spectrum Disorder (FVSD)—historically referred to as Fetal Valproate Syndrome (FVS) or valproic acid embryopathy—is a lifelong, irreversible condition.[9] The severity of the phenotype is highly dose-dependent, with daily maternal doses exceeding 600 mg drastically increasing the teratogenic potential, and polytherapy regimens pushing the risk of major congenital malformations as high as 50%.[28]

## 2.1 Morphological Anomalies: Syndactyly, Craniofacial Dysmorphism, and Skeletal Defects

The physical manifestations of FVSD are profound and identifiable from early infancy. Exposure to sodium valproate during the first trimester critically disrupts the closure of the neural tube, resulting in a high incidence of neural tube defects (NTDs). The occurrence of spina bifida in valproate-exposed neonates is documented to be 20 times higher than in the general, unexposed population.[4] Furthermore, the teratogen induces a highly characteristic pattern of craniofacial dysmorphism. Anthropometric studies of exposed children reveal a significantly increased cephalic index, indicating a disproportionate increase in the width of the skull relative to its anterior-posterior length.[31] Additional dysmorphic features include telecanthus (increased distance between the medial canthi of the eyes), epicanthic folds, a flat nasal bridge with anteverted nostrils, a wide and shallow philtrum, and a thin upper lip with downward-pointing labial commissures.[31]

More strikingly, sodium valproate significantly disrupts the development of the appendicular skeleton. The drug interferes with the median apical ectodermal ridge (AER) signaling during embryonic limb patterning.[33] This disruption precipitates severe musculoskeletal and limb reduction defects, most frequently presenting as cutaneous syndactyly (the abnormal fusion of digits, commonly referred to as "webbed hands"), arachnodactyly (abnormally long and slender digits), clinodactyly, triphalangeal thumbs, radial deviation of the wrists, and, in severe cases, radial ray aplasia and phocomelia.[28] The manifestation of syndactyly and other digit contractures highlights valproate's capacity to arrest apoptosis in the interdigital necrotic zones during the critical window of limb formation, a direct consequence of its pharmacological mechanism of action as an epigenetic modulator.[34] Furthermore, cardiovascular anomalies (such as ventricular septal defects), urogenital malformations including hypospadias, and bilateral cryptorchidism are frequently reported in exposed cohorts.[28]

## 2.2 Neurodevelopmental Alterations and the Architecture of the Prefrontal Cortex

While the macroscopic physical malformations of FVSD are devastating, the neurological devastation wrought by in utero valproate exposure deeply alters the cellular architecture and functionality of the developing brain. Valproic acid functions primarily as a potent inhibitor of histone deacetylases (HDACs).[37] While this epigenetic modulation—which results in transient histone hyperacetylation—is highly effective in suppressing seizure activity and stabilizing mood, it is uniquely destructive during embryogenesis.[38] By inhibiting HDAC, sodium valproate fundamentally alters the cell cycle kinetics of neural progenitor cells (NPCs), leading to aberrant proliferation and differentiation trajectories.[37]

Advanced neuroanatomical studies and magnetic resonance imaging (MRI) analyses of children and animal models exposed to sodium valproate reveal profound structural and transcriptomic changes, particularly concentrated within the prefrontal cortex (PFC).[28] The prefrontal cortex, which governs higher-order executive function, emotional regulation, social behavior, and working memory, exhibits a marked reduction in dendritic spine density and an abnormal increase in the thickness of superficial neocortical layers during early postnatal development.[37] Longitudinal MRI studies in human adolescents and young adults exposed to valproate corroborate these findings, demonstrating accelerated loss in ventral and rostral prefrontal cortex volumes, which correlates directly with cognitive impairment and behavioral dysfunction.[40]

Furthermore, in utero valproate exposure severely disrupts the migration, distribution, and maturation of GABAergic interneurons. Unbiased stereological analyses demonstrate a

significant quantitative loss of parvalbumin-immunoreactive (PV+) and somatostatin-positive (SST+) interneurons within the prefrontal cortex and striatum.[39] Because these interneurons are fundamentally responsible for maintaining the brain's excitatory/inhibitory (E/I) balance, their depletion results in chronic cortical hyperexcitability and neural network instability.[39] This cellular disruption is driven by transcriptomic dysregulation, evidenced by decreased mRNA levels for critical genes such as Pvalb and Kcnc1, the latter of which codes for the voltage-gated potassium channel Kv3.1 specifically expressed in PV+ neurons.[43]
This profound E/I imbalance is compounded by valproate's selective disruption of the mesotelencephalic dopaminergic tracts. Prenatal exposure reduces dopaminergic output from the ventral tegmental area (VTA) to the ventrobasal forebrain and nucleus accumbens (NAc)—regions central to the brain's social reward network.[38] The confluence of prefrontal cortical dysgenesis, interneuron depletion, and dopaminergic disruption provides the specific biological mechanisms underpinning the exceptionally high prevalence of Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD), severe cognitive delay, and expressive language impairment observed in children diagnosed with FVSD.[28]

| **Teratogenic Category** | **Specific Pathologies Linked to in utero Sodium Valproate Exposure** | **Putative Biological Mechanisms** |
|---|---|---|
| **Craniofacial & Organ** | Increased cephalic index, telecanthus, cleft palate, spina bifida, cardiac septal defects. [31] | HDAC inhibition causing aberrant gene transcription and folate metabolism disruption. [28] |
| **Musculoskeletal** | Cutaneous syndactyly (webbed hands), arachnodactyly, radial ray aplasia, joint laxity. [32] | Disruption of apical ectodermal ridge (AER) signaling; arrested interdigital apoptosis. [33] |
| **Neurodevelopmental** | ASD, ADHD, dyspraxia, severe cognitive impairment, accelerated prefrontal cortex volume loss. [28] | Depletion of PV+/SST+ interneurons, E/I imbalance, altered dopaminergic mesotelencephalic tracts. [39] |

# 3. The "Factory Excess" Paradigm and the Expansion of Corporate Markets

The global supply chain for sodium valproate is expansive and highly industrialized. Following the expiration of Sanofi's original patents for the Depakine and Depakote formulations in 1998, the market fractured, allowing a vast network of generic pharmaceutical manufacturers to commoditize the chemical.[49] The historical commercialization of sodium valproate has drawn intense scrutiny, particularly regarding allegations that an overproduction—or a "factory excess"—of the chemical prompted pharmaceutical entities to aggressively pivot their marketing strategies toward off-label psychiatric indications and male demographics to offset the shrinking female epilepsy market.

## 3.1 Industrial Production Scale and the Mourenx Environmental Crisis

The industrial scale required to maintain the global supply of sodium valproate is massive, and the consequences of this high-volume manufacturing have periodically spilled over into environmental crises. This was highlighted starkly at Sanofi's primary manufacturing facility in Mourenx, located in the Pyrenees region of southern France. The Mourenx plant is the central hub for the synthesis of the active pharmaceutical ingredient for Depakine, Depakote, and Depamide.[22]

In 2018, operations at the Mourenx facility were forcibly halted after local environmental groups and independent regulators discovered that the plant was emitting bromopropane—a highly toxic neurotoxin and classified possible carcinogen used as a solvent in the valproate manufacturing process—at levels up to 7,000 times the legal environmental limit.[22] Despite pledges of technical improvements and the installation of activated carbon treatment units, the facility faced renewed scrutiny and was forced to suspend operations again in late 2023 following further toxic leaks. Sanofi attributed the November 2023 breach to the degradation of the carbon filters caused by severe weather, yet the recurring nature of these emissions underscores the immense, relentless volume of chemical processing required to sustain global market demands.[50] The environmental fallout has even led to adjacent civil litigation; in 2023, a woman employed near the plant filed a lawsuit alleging that her two children were born with FVSD-like neurological disorders due exclusively to ambient environmental exposure to the factory's emissions, despite her never ingesting the medication.[50]

## 3.2 The Strategic Pivot to Bipolar Disorder and Male Demographics

As the scientific literature detailing the teratogenic horrors of sodium valproate solidified throughout the 1980s and 1990s, the pharmaceutical industry recognized an impending demographic contraction. To sustain production volumes and profitability, corporate strategy aggressively expanded the drug's licensed and off-label indications.[15] By the late 1990s and early 2000s, valproate derivatives were heavily marketed for the treatment of acute manic episodes associated with bipolar affective disorder, as well as for the prophylaxis of migraine headaches.[5]

This strategic pivot was highly lucrative but fraught with legal malfeasance. In the United States, Abbott Laboratories—the primary distributor of the Depakote brand—faced immense federal scrutiny for illegally pushing the drug beyond its authorized scope. In 2012, Abbott agreed to a staggering $1.6 billion settlement, pleading guilty to criminal and civil violations of the Food, Drug, and Cosmetics Act. The Department of Justice revealed that Abbott had deployed specialized sales forces to illegally market Depakote in nursing homes for the off-label

treatment of elderly dementia patients, and as an add-on therapy for schizophrenia, despite internal clinical trials proving the drug offered no added benefit for these conditions.[23]
In Europe, the expansion into the bipolar and male epilepsy markets effectively insulated the drug's profitability during a period when regulatory scrutiny of female prescribing was escalating. Because the prevailing medical consensus viewed the teratogenic threat exclusively through the lens of maternal exposure in utero, male patients and post-menopausal women became the primary targets for continued valproate expansion.[15] This pivot ensured that factory production lines, such as those in Mourenx, continued to operate at maximum capacity, effectively absorbing any "factory excess" generated by the declining female market. However, the discovery of paternal epigenetic risks in 2023 has shattered this final demographic safe haven, subjecting male prescribing to the same stringent regulatory constraints previously reserved for women.[21]

## 3.3 Corporate Profitability and Distribution Networks in Europe

The modern distribution of sodium valproate across Ireland, England, and France is sustained by a complex web of originator and generic entities.

- **Republic of Ireland:** Sanofi-Aventis Ireland Ltd. remains the principal marketing authorization holder for the Epilim brand. The physical logistics and distribution to retail and hospital pharmacies are managed by major pharmaceutical wholesalers, principally United Drug Wholesale and the Uniphar Group, which dominate the Irish supply chain.[55]
- **England:** The MHRA licenses a vast array of generic valproate formulations alongside Sanofi's proprietary brands. Major distributors supplying the National Health Service (NHS) include Wockhardt UK Ltd, Viatris UK Healthcare, Teva UK Ltd, Kent Pharma, and Zentiva Pharma UK Ltd.[57]
- **France:** Sanofi retains a dominant market share with its Depakine portfolio, though generic distributors such as Laboratoire Aguettant also hold significant distribution profiles.[57]

Despite the overwhelming volume of litigation and reputational damage associated with the valproate scandal, the macro-economic impact on Sanofi's profitability has been largely neutralized by the company's aggressive transition into high-margin biological therapeutics. In its 2024 annual financial reports, Sanofi reported net global sales of €41.1 billion, representing an 11.3% growth at constant exchange rates.[59] This immense profitability is driven almost entirely by its immunology blockbuster Dupixent, which alone generated over €13 billion in sales in 2024, alongside robust performances from newly launched rare disease and oncology medications such as ALTUVIIIO and Ayvakit.[59]
Consequently, sodium valproate—once a flagship revenue generator—now represents a fraction of Sanofi's gross profit margin. The billions of euros generated by Sanofi's biopharma segment allow the corporation to easily absorb legal penalties and defense costs across multiple jurisdictions. This financial reality exacerbates the frustration of patient advocacy groups, who argue that pharmaceutical giants continue to reap record-breaking profits while

aggressively contesting compensation mechanisms for families crippled by their legacy products.[26]

# 4. Current Prescribing Guidelines and Regulatory Frameworks (2024–2025)

The overwhelming epidemiological evidence of sodium valproate's teratogenicity has forced regulatory bodies in Ireland, England, and France to continuously tighten their prescribing protocols. Historically, the medical establishment demonstrated a systemic reluctance to withdraw the drug due to its unparalleled efficacy in controlling treatment-resistant epilepsies and preventing SUDEP.[21] However, the regulatory paradigm shifted dramatically between 2018 and 2025, culminating in some of the strictest prescribing mandates ever applied to an authorized medication.

## 4.1 The Pregnancy Prevention Programme (PPP)

Following a comprehensive safety review by the European Medicines Agency (EMA) in 2018, healthcare regulators across Europe—including the MHRA in the UK, the HPRA in Ireland, and the ANSM in France—instituted the Pregnancy Prevention Programme (PPP).[64] The PPP fundamentally altered the prescribing landscape, mandating that valproate-containing medicines must no longer be used in women or girls of childbearing potential unless the stringent conditions of the programme are met.[66]

The PPP requires prescribers to comprehensively assess female patients for the potential of becoming pregnant. Patients must undergo highly effective, uninterrupted contraception for the entirety of their treatment duration and complete regular pregnancy testing.[64] Crucially, all female patients are required to undergo an annual review with a specialist neurologist or psychiatrist, during which they must sign an Annual Risk Acknowledgement Form (RAF). This form serves as a binding legal and medical document confirming that the patient understands the profound risks of FVSD and the necessary precautions required.[52] Furthermore, the guidelines explicitly state that off-label use of valproate (such as for migraine prophylaxis) is strictly prohibited for women of childbearing potential unless they fully adhere to the PPP, eliminating any regulatory loopholes.[6]

## 4.2 The 2024–2025 Regulatory Pivot: Paternal Exposure and the Under-55 Mandate

In a development that radically redefined the scope of valproate regulation, retrospective observational safety studies published in late 2023 identified a potential risk of neurodevelopmental disorders in children whose fathers were treated with valproate monotherapy in the three months prior to conception.[54] Conducted using electronic medical records from Nordic countries at the behest of the EMA, the study suggested that valproate's capacity to induce testicular toxicity, impaired fertility, and epigenetic alterations in

spermatozoa could result in transgenerational teratogenesis.[21]
In response to this data, the UK's MHRA implemented aggressive new regulatory measures effective from January 31, 2024, with updated safety and educational materials issued in June 2025 and further clarifications published in February 2025.[20] Under these unprecedented rules, sodium valproate must **not** be initiated in *any* new patient—male or female—under the age of 55 unless two independent specialists independently consider and document that no other effective or tolerated treatment exists, or that compelling clinical reasons dictate that the reproductive risks do not apply.[70]
This "two-specialist" sign-off rule represents a massive logistical hurdle designed to systematically reduce valproate initiation. While the dual-specialist authorization is required for all new initiations and for existing female patients continuing therapy, the UK Commission on Human Medicines (CHM) clarified in February 2025 that male patients who are *already* taking valproate do not require a second specialist review to continue their medication, though they must receive comprehensive counseling regarding paternal risks and family planning.[52] This equalization of risk management across both sexes marks a historic departure from prior decades, ending the era where male patients were treated as a demographic safe haven for valproate expansion.

# 5. Investigatory Timelines and Legal Redress: A Comparative Assessment

The speed, transparency, and efficacy with which European states have investigated the sodium valproate scandal vary drastically. France and England initiated investigations significantly faster than the Republic of Ireland, largely due to differences in civil society mobilization, the availability of class-action legal mechanisms, and differing governmental willingness to confront pharmaceutical liability head-on. The metaphor of "investigating a fire" accurately captures the urgency seen in the UK and France, contrasting sharply with the slow-moving, bureaucratic approach observed in Ireland.

## 5.1 France: The Epicenter of Judicial Action and Criminal Indictments

France, as the birthplace of Sanofi and the primary manufacturer of Depakine, moved with relative speed and unparalleled legal aggression once the scale of the crisis became undeniable. Between 1967 and 2016, an estimated 64,100 to 100,000 pregnancies in France were exposed to valproate, resulting in up to 4,100 children born with severe physical malformations and up to 30,400 children suffering neurodevelopmental defects.[15]
Following intense public pressure led by Marine Martin and the patient advocacy group APESAC (Association d'Aide aux Parents d'Enfants souffrant du Syndrôme de l'Anti-Convulsivant), the French social affairs inspection agency (IGAS) launched an investigation, publishing a damning report in February 2016. The report exposed the inertia of both Sanofi and French and European health authorities in failing to act on emerging scientific

knowledge.[26] By 2017, the French government established a state-backed compensation mechanism overseen by the National Office for Indemnification of Medical Accidents (ONIAM) to provide redress to victims.[73]

Simultaneously, APESAC launched the first health-sector class-action lawsuit in French history against Sanofi.[73] The French criminal justice system also intervened aggressively. In 2020, French prosecutors placed Sanofi under formal criminal investigation, bringing charges of aggravated fraud, unintentionally causing injury, and, most severely, **manslaughter** related to the deaths of four infants exposed to the drug.[76]

Sanofi has consistently rebutted these charges and filed legal challenges against the indictments. The corporation argues that it fulfilled its obligations to inform health authorities of fetal malformation risks in the 1980s and neurodevelopmental risks in 2003. Sanofi's defense rests heavily on the assertion that the French Health Agency (ANSM) repeatedly refused the company's requests to update the patient leaflets to explicitly mention congenital malformations and neurodevelopmental delays until 2010, fearing that such warnings would cause pregnant women to abruptly cease treatment, thereby risking lethal maternal seizures.[8]

Despite this defense, the Paris Court issued a historic judgment in January 2022, ruling against Sanofi in the class action. The court determined that Sanofi produced a defective product and failed in its duty of care and information regarding congenital malformations for exposures prior to 2006, and neurodevelopmental disorders for exposures between 2001 and 2006.[74]

## 5.2 England: The Cumberlege Review and Systemic Cultural Autopsy

In the United Kingdom, the response was driven by a sweeping governmental inquiry rather than immediate criminal litigation. In February 2018, the Secretary of State for Health commissioned the Independent Medicines and Medical Devices Safety (IMMDS) Review, chaired by Baroness Julia Cumberlege.[77]

The Cumberlege Report, published in July 2020 under the title *"First Do No Harm,"* investigated the health system's response to sodium valproate, alongside Primodos (hormone pregnancy tests) and pelvic mesh implants.[16] The findings were devastating. Baroness Cumberlege reported an "intensity of suffering" unparalleled in her career, identifying a systemic culture of denial within the healthcare system, an institutional arrogance that dismissed the valid concerns of female patients, and an abject failure by clinicians and the MHRA to collect or act upon safety data.[61] The Review revealed that regulatory bodies were aware of the teratogenic risks of valproate in the 1970s, yet explicitly chose to withhold these warnings from patients to avoid causing "fruitless anxiety".[19]

The publication of the report triggered immediate governmental action. In July 2020, the UK Government issued a fulsome, unreserved apology on behalf of the healthcare system to the affected families.[81] In response to the Review's nine strategic recommendations, the government initiated legislation to appoint an independent Patient Safety Commissioner with statutory responsibilities and established a national Valproate Registry to monitor prescribing adherence and patient outcomes in real-time.[78]

## 5.3 Republic of Ireland: Bureaucratic Inertia and the July 2025 Inquiry

In stark contrast to the swift judicial interventions in France and the comprehensive cultural autopsy in England, the Republic of Ireland's response has been characterized by profound bureaucratic delays and political hesitation. Sodium valproate (Epilim) was licensed in Ireland in 1975, and the Health Service Executive (HSE) estimates that up to 1,250 Irish families have been catastrophically impacted by FVSD over the past five decades.[12]

Despite the Oireachtas Health Committee recommending an inquiry as early as May 2018, and a formal government commitment made by the Minister for Health in November 2020, the inquiry did not officially commence until **July 22, 2025**.[11] The extensive delay between the agreement of the Terms of Reference in July 2023 and the commencement in 2025 was primarily attributed to legal and administrative hurdles surrounding data protection and privacy laws.[17] Because the inquiry handles highly sensitive special category medical data, the government was forced to draft and pass bespoke secondary legislation—namely, S.I. No. 350/2025 and S.I. No. 352/2025 under the Data Protection Act 2018—to establish a legal Joint Controller Agreement between the Minister for Health and the Inquiry Chair, Ms. Bríd O'Flaherty BL.[9]

Furthermore, the Irish inquiry has generated profound controversy due to its strictly **non-statutory** nature. Patient advocacy groups, including OACS Ireland and Epilepsy Ireland, fervently campaigned for a statutory inquiry equipped with robust legal powers to compel witnesses and demand documentation under oath.[84] However, the Irish government insisted on a non-statutory framework. The rationale for this format, while frustrating to campaigners, is rooted in the state's historical experience with public inquiries; non-statutory reviews offer greater procedural flexibility and theoretically faster publication timelines, avoiding the highly adversarial, protracted, and astronomically expensive nature of statutory tribunals (where legal fees frequently consume the majority of the budget and delay findings for years).[61]

The non-statutory Irish inquiry is currently operating across three distinct strands:

1. **Review Phase:** Establishing a definitive timeline of sodium valproate regulation, prescribing, and dispensing in the State from 1975 to the present, documenting the evolution of scientific knowledge regarding teratogenicity.
2. **Personal Statements:** Providing a private, secure forum for individuals diagnosed with FVSD, their mothers, and family members to present oral testimonies detailing their lived experiences.
3. **Health Service Capacity:** Assessing the HSE's current capacity to disseminate and implement safety measures regarding anti-seizure medications, and recommending future service developments and targeted supports for those impacted.[85]

| Jurisdictional Response | Key Investigative Action | Year Commenced | Legal/Statutory Status | Primary Outcomes / |
| --- | --- | --- | --- | --- |

| | | | | Current Status |
|---|---|---|---|---|
| **France** | IGAS Report & Criminal Investigation | 2016 / 2020 | Statutory / Criminal / Civil | Sanofi indicted for manslaughter; 2022 Paris Court ruling against Sanofi; ONIAM State compensation fund active. [74] |
| **England** | IMMDS Review (Cumberlege) | 2018 | Non-Statutory Review | Publication of "First Do No Harm" Report (2020); Unreserved Government Apology; Appointment of Patient Safety Commissioner. [77] |
| **Ireland** | Independent Inquiry (Bríd O'Flaherty BL) | July 2025 | Non-Statutory Inquiry | Currently active. Delayed by Data Protection legislation (S.I. 352/2025). Focus on historical timeline and personal oral statements. [47] |

# 6. The Emergence of Medical Cannabis Oil: Clinical Paradigms and Economic Disparities

As the prescription of sodium valproate is systematically restricted and phased out for vast segments of the epileptic population, neurologists and patients are urgently seeking alternative, non-teratogenic therapies for severe, treatment-resistant epilepsy. In recent years, medical cannabis—specifically highly purified cannabidiol (CBD) isolates—has emerged as a highly publicized and sought-after therapeutic alternative. However, transitioning a patient from a traditional, broad-spectrum AED like valproate to a cannabinoid regimen introduces complex challenges regarding clinical efficacy, severe pharmacological interactions, and profound economic disparities.

## 6.1 Clinical Efficacy: CBD versus Sodium Valproate

Historically, the medical establishment viewed cannabis oil with deep skepticism due to a lack of robust, randomized controlled trials (RCTs) and the proliferation of unregulated artisanal products.[88] However, this paradigm shifted definitively with the FDA and EMA approval of Epidyolex, a pharmaceutical-grade, 99% pure CBD oral solution. High-powered, double-blind, placebo-controlled RCTs have provided Class 1 evidence that the adjunctive use of CBD significantly improves seizure control in specific, catastrophic childhood epilepsy syndromes—namely Lennox-Gastaut syndrome (LGS), Dravet syndrome (DS), and Tuberous Sclerosis Complex (TSC).[88]

In pivotal phase-3 clinical trials for LGS and Dravet syndrome, patients receiving 20mg/kg/day of CBD experienced an approximate 43.9% reduction in the frequency of drop seizures compared to baseline, drastically outperforming the placebo cohorts.[90] Furthermore, retrospective observational studies of full-spectrum medical cannabis extracts (containing both CBD and specific ratios of THC, such as a 20:1 ratio) have reported up to a 97% mean reduction in monthly seizure frequency in highly intractable pediatric cases where traditional AEDs completely failed.[25]

However, a critical distinction must be drawn. Sodium valproate is a broad-spectrum anticonvulsant highly effective across almost all generalized and focal seizure types.[4] Conversely, the clinically proven efficacy of CBD remains narrowly restricted to rare genetic syndromes. Neurological societies, including the British Paediatric Neurology Association (BPNA) and Epilepsy Ireland, explicitly warn that the positive data derived from LGS/DS trials cannot be automatically extrapolated to assume that CBD will be an effective monotherapy or adjunctive treatment for the broader adult epileptic population.[88]

## 6.2 Pharmacological Interactions and the Hepatotoxicity Risk Profile

A critical complexity in substituting or combining medical cannabis with sodium valproate lies in their highly volatile pharmacokinetic interactions. CBD is heavily metabolized by the liver and acts as a potent inhibitor of several cytochrome P450 enzymes (specifically CYP2C19 and CYP3A4).[94]

When CBD is administered alongside existing AEDs, it dramatically alters their plasma concentrations and metabolic pathways. For example, CBD significantly increases the levels of N-desmethylclobazam (the active metabolite of the benzodiazepine clobazam), intensifying sedative side effects to the point of severe somnolence.[91] More alarmingly, clinical and pharmacovigilance data indicate a severe, bidirectional interaction between CBD and sodium valproate. The co-administration of CBD and VPA is strongly associated with a highly elevated risk of hepatotoxicity, manifesting as significant, potentially dangerous spikes in liver transaminases (AST and ALT).[95] Consequently, clinicians attempting to transition patients from valproate to CBD must navigate a treacherous pharmacological landscape, requiring intense hepatic monitoring at baseline, two weeks post-initiation, and following any dose escalation.[96]

Furthermore, while CBD avoids the morphological horrors of FVSD, it is not a benign substance. Common adverse drug reactions (ADRs) associated with pharmaceutical CBD include profound drowsiness, diarrhea, decreased appetite, vomiting, and ataxia.[88] Additionally, pediatric and neurological societies issue stark warnings against the use of unregulated, full-spectrum cannabis oils containing high concentrations of Tetrahydrocannabinol (THC). The benefits of THC in treating epilepsy have not been scientifically established, and its psychoactive properties carry severe risks of impairing neurodevelopment and causing long-term cognitive damage in developing pediatric brains.[89] Paradoxically, some pharmacovigilance reports have even highlighted that unregulated CBD use can precipitate aggravations of epilepsy and status epilepticus.[99]

## 6.3 Legal Frameworks, Cost Disparities, and Market Accessibility

The transition from sodium valproate to medical cannabis is ultimately bottlenecked by convoluted legal frameworks and extreme financial barriers.

Sodium valproate, having been off-patent for decades, is a highly commoditized generic medication. The cost to national healthcare systems and individual patients is exceptionally low, ensuring ubiquitous, barrier-free accessibility across all socioeconomic strata.[49] Conversely, medical cannabis occupies an elite pricing tier, creating a profound class disparity in advanced epilepsy care.

In the Republic of Ireland, the government established the Medical Cannabis Access Programme (MCAP) in 2019 to permit medical consultants to legally prescribe specific, approved cannabis-based products for a highly restricted set of conditions, including severe, treatment-refractory epilepsy.[24] However, the logistical and financial hurdles to access this programme are immense. General Practitioners (GPs) are legally prohibited from prescribing these products; patients must undergo rigorous assessment by a hospital consultant, who must then apply for a bespoke 'Ministerial Licence' from the Department of Health.[24]

Because the domestic supply chain for medical cannabis remains embryonic, products are frequently imported from international dispensaries, such as the Transvaal Apotheek in the Netherlands.[24] The financial burden of these imports is staggering. Private prescriptions for medical cannabis oils can average over €186 per month for basic preparations, while full-spectrum imports and complex cannabinoid therapies can cost desperate families between £1,800 and €2,000 per month.[24] While the HSE's Primary Care Reimbursement Service (PCRS) provides funding for patients successfully approved under the MCAP, the stringent clinical inclusion criteria and bureaucratic friction leave the vast majority of the epileptic population locked out of the programme.[24] Consequently, countless patients remain reliant on traditional, highly toxic AEDs like valproate, or are forced to navigate the legal and clinical hazards of illicit, unregulated cannabis markets.[24]

| **Therapeutic Profile** | **Sodium Valproate (Epilim, Depakine, Depakote)** | **Medical Cannabis & CBD (Epidyolex, Full-Spectrum Oils)** |
| --- | --- | --- |
| **Clinical Efficacy** | Broad-spectrum. Highly effective for generalized/focal | Narrow-spectrum. Proven Class 1 efficacy for rare genetic |

| | | |
|---|---|---|
| | seizures, bipolar disorder, and migraine prophylaxis. [4] | syndromes (LGS, Dravet, TSC). [88] |
| **Teratogenic Risk** | Extremely High. 10% risk of congenital malformations; 40% risk of neurodevelopmental disorders (FVSD). [12] | Unknown/Low teratogenicity. However, THC carries high risks for postnatal neurocognitive development. [89] |
| **Adverse Drug Reactions** | Weight gain, alopecia, testicular toxicity, profound embryonic disruption. [21] | Somnolence, diarrhea, ataxia, decreased appetite, potential seizure aggravation. [90] |
| **Hepatotoxicity (Liver Risk)** | Moderate to High. [95] | High, specifically when co-administered with Valproate. Causes severe AST/ALT elevation. [96] |
| **Regulatory & Legal Status** | Heavily restricted. Initiation banned for all <55s without dual-specialist authorization. [21] | Highly restricted. Requires Consultant-led Ministerial Licence and MCAP approval in Ireland. [24] |
| **Cost & Market Access** | Widely accessible, cheap generic medication. Sustained by major wholesale distributors. [49] | Exceedingly high cost (€186 to >€2000/month). Subject to severe reimbursement friction via PCRS. [24] |

# 7. Conclusions

The continued prescription of sodium valproate occupies one of the most intensely contested intersections of neuropharmacology, medical ethics, and corporate jurisprudence. The drug's historically unparalleled efficacy in suppressing lethal epileptic seizures and stabilizing psychiatric disorders is permanently overshadowed by its epigenetic capacity to induce Fetal Valproate Spectrum Disorder. By inhibiting histone deacetylase, valproate fundamentally alters the trajectory of embryonic development, destroying the cellular architecture of the prefrontal cortex, depleting vital interneuron populations, and arresting the morphological formation of the appendicular skeleton.

The investigatory and legal responses across Europe reveal a profound disparity in state accountability and systemic agility. France, the manufacturing locus of the drug, utilized robust criminal indictments and civil class-action mechanisms to successfully challenge Sanofi, establishing comprehensive state compensation frameworks. England engaged in a systemic cultural autopsy via the Cumberlege Review, exposing an institutional arrogance that ignored patient harm, which fundamentally altered its patient safety legislation. In stark contrast, the Republic of Ireland's response has been defined by chronic, agonizing bureaucratic inertia. The delayed commencement of the non-statutory inquiry in July 2025—stalled by complex data protection legislation and political resistance to the financial burdens of a statutory tribunal—highlights a systemic reluctance to rapidly address iatrogenic harm, leaving over a

thousand affected families waiting decades for formal recognition.
Furthermore, the pharmaceutical strategy surrounding valproate demonstrates the ruthless agility of corporate market forces. As the female demographic was rightfully restricted due to teratogenicity warnings, manufacturers expanded the drug's footprint into bipolar disorder and male cohorts, necessitating massive, environmentally hazardous industrial outputs that resulted in severe crises like the Mourenx bromopropane leaks. The 2024–2025 regulatory mandates, which finally acknowledge the epigenetic risks of paternal exposure, close this historical loophole, subjecting both sexes under the age of 55 to unprecedented dual-specialist oversight and effectively ending valproate's reign as a first-line therapy.
As valproate faces regulatory obsolescence for younger demographics, medical cannabis represents a vital, albeit highly complicated, therapeutic frontier. While pharmaceutical-grade CBD isolates offer proven efficacy for severe genetic epilepsies without the teratogenic horrors of FVSD, their clinical integration is severely hampered by extreme financial costs, highly restrictive access programs such as Ireland's MCAP, and dangerous hepatotoxic interactions with legacy AEDs. Ultimately, until national healthcare systems can bridge the profound economic and regulatory gap between cheap, toxic generics and expensive, highly-regulated cannabinoids, patients and clinicians will remain trapped, forced to navigate the treacherous margins between achieving seizure control and guaranteeing patient safety.

## Works cited


1. Sodium Valproate Timeline – Key Events July 2020, accessed on March 19, 2026, https://www.immdsreview.org.uk/downloads/Annexes/Annex-C-Valproate-timeline.pdf
2. The Independent Medicines and Medical Devices Safety Review Written Evidence Manufacturers of Sodium Valproate, accessed on March 19, 2026, https://www.immdsreview.org.uk/downloads/Evidence/FOR%20PUBLICATION%20-%20Manufacturers%20of%20Sodium%20Valproate.pdf
3. SECOND SUPPLEMENT DATED 27 FEBRUARY 2026 TO THE BASE PROSPECTUS DATED 4 JUNE 2025 - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/pages/docs/investor-relations/fixed-income/base-prospectus/Sanofi_EMTN-programme_2025-2nd-supplement---visa-26-040.pdf
4. Awareness and documentation of the teratogenic effects of valproate among women of child-bearing potential - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6465217/
5. Annual Report on Form 20-F - SEC.gov, accessed on March 19, 2026, https://www.sec.gov/Archives/edgar/data/1121404/000119312507072848/d20f.htm
6. Shared Care Protocol Valproate medicines for all patients (male or female) for all indications, accessed on March 19, 2026, https://nhssomerset.nhs.uk/wp-content/uploads/sites/2/SCP-Valproate-v1.3-March-2025-FINAL.pdf
7. 2022 Sanofi 20-F, accessed on March 19, 2026,

https://www.sanofi.com/assets/dotcom/content-app/publications/annual-report-on-form-20-f/2022-01-01-form-20-f-2022-en.pdf
8. Regarding the ongoing Depakine® (sodium valproate) product litigation in France, Sanofi is issuing the following statement, con, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/pages/docs/our-company/social-impact/responsible-business-values/information-on-sodium-valproate/20220830_statement-on-depakine.pdf
9. FAQs | Sodium Valproate Inquiry Ireland, accessed on March 19, 2026, https://sodiumvalproateinquiry.ie/faq/
10. Fetal valproate spectrum disorder - Wikipedia, accessed on March 19, 2026, https://en.wikipedia.org/wiki/Fetal_valproate_spectrum_disorder
11. Valproate: 'Who knew what and when?' - Medical Independent, accessed on March 19, 2026, https://www.medicalindependent.ie/in-the-news/news-features/valproate-who-knew-what-and-when/
12. Epilepsy Ireland make Sodium Valproate Inquiry Submission, accessed on March 19, 2026, https://www.epilepsy.ie/content/epilepsy-ireland-make-sodium-valproate-inquiry-submission
13. Sanofi faces formal investigation over epilepsy drug Depakine - Pharmaceutical Technology, accessed on March 19, 2026, https://www.pharmaceutical-technology.com/news/sanofi-depakine-formal-investigation/
14. Minister for Health provides update on Sodium Valproate inquiry - Epilepsy Ireland, accessed on March 19, 2026, https://www.epilepsy.ie/content/minister-health-provides-update-sodium-valproate-inquiry
15. Sanofi epilepsy drug linked to over 4000 child deformities - The Guardian, accessed on March 19, 2026, https://www.theguardian.com/society/2017/apr/21/sanofi-epilepsy-drug-linked-to-over-4000-child-deformities
16. The Baroness Cumberlege Report – First do no harm (Published 8th July 2020) - HQIP, accessed on March 19, 2026, https://www.hqip.org.uk/the-baroness-cumberlege-report-first-do-no-harm-published-8th-july-2020/
17. 'The real work starts now': Long-awaited inquiry into drug that causes birth defects begins, accessed on March 19, 2026, https://www.thejournal.ie/inquiry-sodium-valproate-antiepilepsy-drug-birth-defects-6770258-Jul2025/
18. Departmental Inquiries – Thursday, 26 Feb 2026 – Parliamentary Questions (34th Dáil) – Houses of the Oireachtas, accessed on March 19, 2026, https://www.oireachtas.ie/en/debates/question/2026-02-26/492/
19. The Journal News Article: Campaigners seek to meet Minister as ..., accessed on March 19, 2026, https://oacsireland.com/the-journal-news-article-campaigners-seek-to-meet-mi

nister-as-inquiry-into-drug-that-causes-birth-defects-yet-to-begin/
20. Valproate (Belvo, Convulex, Depakote, Dyzantil, Epilim, Epilim Chrono or Chronosphere, Episenta, Epival, and Syonell ): new safety and educational materials to support regulatory measures in men and women under 55 years of age - GOV.UK, accessed on March 19, 2026, https://www.gov.uk/drug-safety-update/valproate-belvo-convulex-depakote-dyzantil-epilim-epilim-chrono-or-chronosphere-episenta-epival-and-syonellv-new-safety-and-educational-materials-to-support-regulatory-measures-in-men-and-women-under-55-years-of-age
21. Guidance for Valproate prescribing in Adult (16 and over) Neurology - Association of British Neurologists, accessed on March 19, 2026, https://cdn.ymaws.com/www.theabn.org/resource/collection/6750BAE6-4CBC-4DDB-A684-116E03BFE634/2025-06-12-ABN_Guidance_Valproate_update_June25-withFAQs.pdf
22. Sanofi halts production at Pyrenees plant after pollution complaint - FirstWord Pharma, accessed on March 19, 2026, https://firstwordpharma.com/story/4618485
23. Sanofi faces French charges in Depakine birth defects probe - Fierce Pharma, accessed on March 19, 2026, https://www.fiercepharma.com/pharma/sanofi-facing-french-charges-depakine-birth-defects-probe
24. Medical Cannabis Products Reimbursement Scheme - HSE, accessed on March 19, 2026, https://www2.hse.ie/services/schemes-allowances/medical-cannabis-products-reimbursement-scheme/
25. Ending the pain of children with severe epilepsy? An audit of the impact of medical cannabis in 10 patients - ResearchGate, accessed on March 19, 2026, https://www.researchgate.net/publication/347569352_Ending_the_pain_of_children_with_severe_epilepsy_An_audit_of_the_impact_of_medical_cannabis_in_10_patients
26. The Independent Medicines and Medical Devices Safety Review Written Evidence Patient Groups: Sodium Valproate, accessed on March 19, 2026, https://www.immdsreview.org.uk/downloads/Evidence/FOR%20PUBLICATION%20-%20Patient%20Groups%20-%20Sodium%20Valproate.pdf
27. Fetal Valproate Syndrome with Limb Defects: An Indian Case Report - PMC - NIH, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5143702/
28. Valproic Acid in Pregnancy Revisited: Neurobehavioral, Biochemical and Molecular Changes Affecting the Embryo and Fetus in Humans and in Animals: A Narrative Review - MDPI, accessed on March 19, 2026, https://www.mdpi.com/1422-0067/25/1/390
29. OACS Ireland, accessed on March 19, 2026, https://oacsireland.com/
30. Valproic Acid in Pregnancy Revisited: Neurobehavioral, Biochemical and Molecular Changes Affecting the Embryo and Fetus in Humans and in Animals: A Narrative Review - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10779436/
31. Exposure to sodium valproate during pregnancy: Facial features and signs of

autism - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5555794/
32. Congenital Malformations Associated with Maternal Use of Valproic Acid - Cambridge University Press & Assessment, accessed on March 19, 2026, https://www.cambridge.org/core/services/aop-cambridge-core/content/view/31A67BB3058D0EA0A96CC51791AD8AF3/S0317167100026639a.pdf/congenital-malformations-associated-with-maternal-use-of-valproic-acid.pdf
33. Split-hand/split-foot malformation associated with maternal valproate consumption, accessed on March 19, 2026, https://utoronto.scholaris.ca/bitstreams/470411d7-06bd-403a-a6b6-9cfc106c0181/download
34. Prenatal exposure to valproic acid during pregnancy and limb deficiencies: A case-control study | Request PDF - ResearchGate, accessed on March 19, 2026, https://www.researchgate.net/publication/12610093_Prenatal_exposure_to_valproic_acid_during_pregnancy_and_limb_deficiencies_A_case-control_study
35. Neurodevelopmental Effects of Fetal Antiepileptic Drug Exposure - PMC - NIH, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4376625/
36. Sodium Valproate Timeline – Key Events Working Draft – April 2020, accessed on March 19, 2026, https://www.immdsreview.org.uk/downloads/Draft_sodium_valproate_timeline_April_2020.pdf
37. In Utero Exposure to Valproic Acid Induces Neocortical Dysgenesis via Dysregulation of Neural Progenitor Cell Proliferation/Differentiation | Journal of Neuroscience, accessed on March 19, 2026, https://www.jneurosci.org/content/36/42/10908
38. Gestational Exposure to Sodium Valproate Disrupts Fasciculation of the Mesotelencephalic Dopaminergic Tract, With a Selective Reduction of Dopaminergic Output From the Ventral Tegmental Area - Frontiers, accessed on March 19, 2026, https://www.frontiersin.org/journals/neuroanatomy/articles/10.3389/fnana.2020.00029/full
39. Prenatal Valproic Acid Exposure Impairs Offspring Cognition Through Disturbing Interneuron Development - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11866047/
40. Preliminary evidence for progressive prefrontal abnormalities in adolescents and young adults with bipolar disorder - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2852397/
41. Altered cortical thickness following prenatal sodium valproate exposure - PMC - NIH, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4184779/
42. Sodium valproate use is associated with reduced parietal lobe thickness and brain volume, accessed on March 19, 2026, https://www.neurology.org/doi/10.1212/WNL.0b013e318292a2e5
43. Prenatal Valproate Exposure Differentially Affects Parvalbumin-Expressing Neurons and Related Circuits in the Cortex and Striatum of Mice - Frontiers,

accessed on March 19, 2026, https://www.frontiersin.org/journals/molecular-neuroscience/articles/10.3389/fnmol.2016.00150/full
44. Alterations of the Hippocampal Networks in Valproic Acid-Induced Rat Autism Model, accessed on March 19, 2026, https://www.frontiersin.org/journals/neural-circuits/articles/10.3389/fncir.2022.772792/full
45. Gestational Exposure to Sodium Valproate Disrupts Fasciculation of the Mesotelencephalic Dopaminergic Tract, With a Selective Reduction of Dopaminergic Output From the Ventral Tegmental Area - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7290005/
46. First Do No Harm – What you need to know about the Cumberlege Review - Bevan Brittan, accessed on March 19, 2026, https://www.bevanbrittan.com/insights/articles/2020/first-do-no-harm-what-you-need-to-know-about-the-cumberlege-review/
47. S.I. No. 352/2025 - Data Protection Act 2018 (Section 60(6)) (Inquiry into the Licensing and Use of Sodium Valproate in Women of Child-Bearing Potential in the State) Regulations 2025 - Irish Statute Book, accessed on March 19, 2026, https://www.irishstatutebook.ie/eli/2025/si/352/made/en/print
48. France's baby steps towards becoming a group litigation forum – the 'Sanofi' ruling (5th January 2022 - N°RG 17/07001) - Pogust Goodhead, accessed on March 19, 2026, https://pogustgoodhead.com/opinions/frances-baby-steps-towards-becoming-a-group-litigation-forum-the-sanofi-ruling-5th-january-2022-nrg-17-07001/
49. Sanofi faces manslaughter charges in France over yearslong Depakine probe, accessed on March 19, 2026, https://www.fiercepharma.com/pharma/sanofi-faces-manslaughter-charges-france-over-yearslong-depakine-probe
50. Troubled Sanofi plant leaked dangerous neurotoxin last month, accessed on March 19, 2026, https://www.fiercepharma.com/manufacturing/after-leak-neurotoxin-sanofi-plant-france-under-scrutiny-again
51. Form S-1 - SEC.gov, accessed on March 19, 2026, https://www.sec.gov/Archives/edgar/data/1423824/000119312514450716/d835750ds1.htm
52. Valproate Policy, accessed on March 19, 2026, https://www.elft.nhs.uk/sites/default/files/2025-06/valproate_policy_3.0.docx
53. New valproate regulations, informed choice and seizure risk - PMC - NIH, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11319429/
54. Sanofi reports issues around safety study into valproate - Epilepsy Society, accessed on March 19, 2026, https://epilepsysociety.org.uk/news/valproate-safety-study-concerns
55. Medicines Shortages Update - IPU - Irish Pharmacy Union, accessed on March 19, 2026, https://ipu.ie/pharmacy-support/ipu-product-file/medicines-shortages/
56. Primary Care Reimbursement Service - Statistical Analysis of Claims and

Payments 2014, accessed on March 19, 2026, https://www.drugsandalcohol.ie/25317/1/PCRS_statAnalyis_2014.pdf
57. valproic acid, sodium valproate, valproate pivoxil, valproate semisodium, valpromide, valproate bismuth, calcium valproate, valp - European Medicines Agency (EMA), accessed on March 19, 2026, https://www.ema.europa.eu/en/documents/psusa/valproic-acid-sodium-valproate-valproate-pivoxil-valproate-semisodium-valpromide-valproate-bismuth-calcium-valproate-valproate-magnesium-list-nationally-authorised-medicinal-products_en.pdf
58. Suppliers for Sodium valproate 500mg gastro-resistant tablets, accessed on March 19, 2026, https://dmd-browser.nhsbsa.nhs.uk/vmp/view/1701/suppliers?ref=
59. Q4 sales growth of 10.3%, 2024 business EPS guidance exceeded, and strong business EPS rebound expected in 2025 - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/pressreleases/2025/2025-01-30-06-30-00-3017713-en.pdf
60. 2025: strong sales and EPS growth. Continued profitable growth expected in 2026 | Sanofi, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/pressreleases/2026/2026-01-29-06-30-00-3228191-en.pdf
61. First Do No Harm – The report of the Independent Medicines and Medical Devices Safety Review, accessed on March 19, 2026, https://www.immdsreview.org.uk/downloads/IMMDSReview_Web.pdf
62. Public Hearing on Valproate - Written interventions - European Medicines Agency (EMA), accessed on March 19, 2026, https://www.ema.europa.eu/en/documents/other/public-hearing-valproate-written-interventions_en.pdf
63. Sodium valproate restrictions to begin in January - Epilepsy Action, accessed on March 19, 2026, https://www.epilepsy.org.uk/news/sodium-valproate-restrictions-to-begin-in-january
64. Valproate in pregnancy and conception - Royal College of Psychiatrists, accessed on March 19, 2026, https://www.rcpsych.ac.uk/mental-health/treatments-and-wellbeing/valproate-in-pregnancy-and-conception
65. Valproate (Epilim) guidelines - Epilepsy Ireland, accessed on March 19, 2026, https://www.epilepsy.ie/content/valproate-epilim-guidelines
66. Valproate medicines (Epilim , Depakote ): Pregnancy Prevention Programme materials online - GOV.UK, accessed on March 19, 2026, https://www.gov.uk/drug-safety-update/valproate-medicines-epilim-depakote-pregnancy-prevention-programme-materials-online
67. Valproate – reproductive risks - GOV.UK, accessed on March 19, 2026, https://www.gov.uk/guidance/valproate-reproductive-risks
68. Precautionary measures for male patients using valproate-containing medicines - HPRA, accessed on March 19, 2026, https://www.hpra.ie/safety-information/how-we-monitor-safety/medicines/use-of

-medicines-in-pregnancy/valproate-(epilim)-use-in-male-patients
69. 1/1 Paris, June 16, 2023. The health of all those treated and impacted by our medicines is our priority. A post-authorisation s - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/pages/docs/our-company/social-impact/responsible-business-values/information-on-sodium-valproate/depakine-website-statement-june-2023-en.pdf
70. January 2025 (updated) Guidance note for the safe prescribing of valproate, accessed on March 19, 2026, https://primarycare.northeastlondon.icb.nhs.uk/wp-content/uploads/2025/04/January-2025-updated-Guidance-note-for-the-safe-prescribing-of-valproate-v-1.1-merged.pdf
71. DRUG SAFETY UPDATE (DSU) - GOV.UK, accessed on March 19, 2026, https://assets.publishing.service.gov.uk/media/67ae07036e6c8d18118acd83/Valproate_DSU_February_2025_-_FINAL_PDF.pdf
72. Drugmaker Sanofi Ordered to Compensate French Family for Epilepsy Drug Side Effects, accessed on March 19, 2026, https://www.insurancejournal.com/news/international/2022/05/16/667786.htm
73. Information on Sodium Valproate - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/en/our-responsibility/sodium-valproate
74. Historic judgment rendered by the Paris Court in a French class action | BCLP, accessed on March 19, 2026, https://www.bclplaw.com/en-US/events-insights-news/historic-judgment-rendered-by-the-paris-court-in-a-french-class-action.html
75. France's Sanofi charged with manslaughter in birth defects case - Al Jazeera, accessed on March 19, 2026, https://www.aljazeera.com/economy/2020/8/3/frances-sanofi-charged-with-manslaughter-in-birth-defects-case
76. Solicitor representing Sodium Valproate patient groups comments on Sanofi charged with manslaughter - Leigh Day, accessed on March 19, 2026, https://www.leighday.co.uk/news/press-releases/2020-news/solicitor-representing-sodium-valproate-patient-groups-comments-on-sanofi-charged-with-manslaughter/
77. the Independent Medicines and Medical Devices safety Review ..., accessed on March 19, 2026, https://www.immdsreview.org.uk/report.html
78. The Cumberlege Report “First do no harm”: 12 months later, the government responds, accessed on March 19, 2026, https://www.dacbeachcroft.com/en/What-we-think/The-Cumberlege-Report-First-do-no-harm-12-months-later-the-government-responds
79. Cumberlege review exposes stubborn and dangerous flaws in healthcare - The BMJ, accessed on March 19, 2026, https://www.bmj.com/content/370/bmj.m3099
80. The Pharmaceutical Year That Was, 2020 - PMC - NIH, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7722252/
81. Update on the government's response to the Independent Medicines ..., accessed on March 19, 2026,

https://www.gov.uk/government/speeches/update-on-the-governments-response-to-the-independent-medicines-and-medical-devices-safety-review
82. Independent Medicines and Medical Devices Safety Review: update report on government implementation - GOV.UK, accessed on March 19, 2026, https://www.gov.uk/government/publications/independent-medicines-and-medical-devices-safety-review-update-report-on-government-implementation/independent-medicines-and-medical-devices-safety-review-update-report-on-government-implementation
83. Minister for Health to establish inquiry into historical licensing & use of Sodium Valproate in pregnancy | Epilepsy Ireland, accessed on March 19, 2026, https://www.epilepsy.ie/content/minister-health-establish-inquiry-historical-licensing-use-sodium-valproate-pregnancy
84. Government approval for inquiry into historical use of sodium valproate - Irish Pharmacist, accessed on March 19, 2026, https://irishpharmacist.ie/news/government-approval-for-inquiry-into-historical-use-of-sodium-valproate/
85. Minister for Health welcomes the commencement of Sodium Valproate Inquiry, accessed on March 19, 2026, https://www.gov.ie/en/department-of-health/press-releases/minister-for-health-welcomes-the-commencement-of-sodium-valproate-inquiry/
86. Sodium valproate inquiry begins after two-year wait - OACS Ireland, accessed on March 19, 2026, https://oacsireland.com/sodium-valproate-inquiry-begins-after-two-year-wait/
87. Sodium Valproate Inquiry Update (2025) - Epilepsy Ireland, accessed on March 19, 2026, https://www.epilepsy.ie/content/sodium-valproate-inquiry-update-2025
88. Evidence in Epilepsy, accessed on March 19, 2026, https://www.epilepsy.ie/content/evidence-epilepsy
89. Our Position on CBD & Medical Cannabis in Epilepsy, accessed on March 19, 2026, https://www.epilepsy.ie/content/our-position-cbd-medical-cannabis-epilepsy
90. Medicinal cannabis and epilepsy: Can CBD help with seizures? | Ora Pharm, accessed on March 19, 2026, https://orapharm.co.nz/medicinal-cannabis-and-epilepsy/
91. Cannabinoids in the Treatment of Epilepsy: Hard Evidence at Last? - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5767492/
92. Cost-effectiveness of cannabidiol (Epidyolex®) for use as an adjunctive therapy of seizures associated with Dravet syndrome, in - National Centre for Pharmacoeconomics, accessed on March 19, 2026, https://www.ncpe.ie/wp-content/uploads/2021/03/Technical-Summary_Cannabidiol-for-DS-HTA-ID-20004-Final-Report.pdf
93. Commentary and critique on the BPNA publication: - The Medical Cannabis Clinicians Society, accessed on March 19, 2026, https://www.ukmccs.org/wp-content/uploads/2021/12/Commentary-and-critique-BNPAs-Guidance-on-the-use-of-cannabis-based-products-for-medicinal-use-in-epilepsy-Oct-2021-1.1.pdf
94. Clinical Guidance on Cannabis for Medical Use, accessed on March 19, 2026,

https://www.drugsandalcohol.ie/38201/1/Clinical%20Guidance%20on%20Cannabis%20for%20Medical%20Use.pdf
95. Cannabidiol: A Review of Clinical Efficacy and Safety in Epilepsy - PubMed, accessed on March 19, 2026, https://pubmed.ncbi.nlm.nih.gov/31053391/
96. Guidance on the use of cannabis-based products for medicinal use in children and young people with epilepsy - British Paediatric Neurology Association, accessed on March 19, 2026, https://bpna.org.uk/userfiles/BPNA_CBPM_Guidance_Oct2018.pdf
97. Practical approach to the safe use of cannabidiol in patients with refractory epilepsy: a mini review - Frontiers, accessed on March 19, 2026, https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2025.1681815/full
98. Cannabis-derived medicinal products - questions and answers - RCPCH, accessed on March 19, 2026, https://www.rcpch.ac.uk/resources/cannabis-derived-medicinal-products-questions-answers
99. Cannabinoids and Adverse Convulsive Effects: A Pharmacovigilance and Addictovigilance Analysis of Cases Reported in France - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12200219/
100. Press release: Q3: continued sales and earnings progress - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/en/media-room/press-releases/2025/2025-10-24-05-30-00-3172594
101. Neurologists' perspectives of cannabis-based medicines: results from an all-Ireland survey, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12031982/
102. Pricing - MyAccess Clinics, accessed on March 19, 2026, https://myaccessclinics.ie/pricing/
103. Composition and Use of Cannabis Extracts for Childhood Epilepsy in the Australian Community - PMC, accessed on March 19, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6033872/
104. Observation of use study with Depakine Chrono Investigator(s) - Sanofi, accessed on March 19, 2026, https://www.sanofi.com/assets/dotcom/content-app/clinical-studies/pharma/Letter-V/L_9557_summary.pdf